

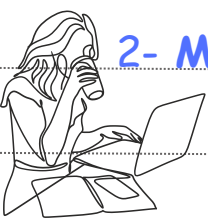


QUESTIONS

- 1- **1- MESSAGE:** THE INFORMATION OR DATA TO BE COMMUNICATED (TEXT, NUMBERS, IMAGES, AUDIO, VIDEO).
- 2- **2- SENDER:** THE DEVICE THAT SENDS THE DATA MESSAGE (COMPUTER, TELEPHONE, VIDEO CAMERA).
- 3- **3- RECEIVER:** THE DEVICE THAT RECEIVES THE MESSAGE (COMPUTER, TV, TELEPHONE).
- 4- **4- TRANSMISSION MEDIUM:** THE PHYSICAL PATH THE MESSAGE TRAVELS (TWISTED-PAIR WIRE, COAXIAL CABLE, FIBER-OPTIC CABLE, RADIO WAVES).
- 5- **5- PROTOCOL:** A SET OF RULES GOVERNING DATA COMMUNICATIONS TO ENSURE DEVICES CAN COMMUNICATE EFFECTIVELY.

- 2- **--> MULTIPOINT CONNECTION** SHARES ONE CONNECTION AMONG DEVICES, REDUCING CABLING COST AND COMPLEXITY.
 --> IT IS MORE ROBUST BECAUSE MULTIPLE DEVICES SHARE THE ,
 SO IF ONE DEVICE FAILS, OTHERS CAN CONTINUE TO COMMUNICATE
 --> IN CONTRAST, POINT-TO-POINT CONNECTIONS DIRECTLY
 DEVICES, REQUIRING MORE CABLING FOR MULTIPLE DEVICES.

- 3- **1- POINT-TO-POINT:** DIRECT CONNECTION BETWEEN TWO DEVICES
- 2- **MULTIPOINT:** ONE COMMUNICATION LINE SHARED AMONG MULTIPLE DEVICES.





QUESTIONS

4- **HALF-DUPLEX:** COMMUNICATION CAN OCCUR IN BOTH DIRECTIONS, BUT NOT AT THE SAME TIME (E.G., WALKIE-TALKIES).

FULL-DUPLEX: COMMUNICATION CAN OCCUR SIMULTANEOUSLY IN BOTH DIRECTIONS (E.G., TELEPHONE CONVERSATIONS).

5- **MESH TOPOLOGY:** $\frac{N(N-1)}{2}$ LINKS (EACH DEVICE CONNECTS TO EVERY OTHER DEVICE).

RING TOPOLOGY: NN LINKS (EACH DEVICE CONNECTED TO TWO OTHERS FORMING A RING).

BUS TOPOLOGY: 1 MAIN CABLE (BACKBONE) SHARED BY ALL DEVICES

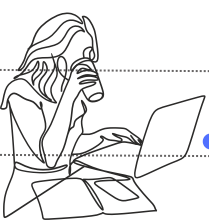
STAR TOPOLOGY: NN LINKS (EACH DEVICE CONNECTED TO A CENTRAL HUB).

6- **A. FIVE DEVICES ARRANGED IN A MESH TOPOLOGY:**

- FAILURE OF ONE CONNECTION USUALLY DOES NOT DISRUPT THE NETWORK BECAUSE MULTIPLE PATHS EXIST BETWEEN DEVICES. NETWORK IS HIGHLY FAULT TOLERANT.

B. FIVE DEVICES ARRANGED IN A STAR TOPOLOGY (NOT COUNTING THE:

- FAILURE OF A DEVICE'S CONNECTION AFFECTS ONLY THAT DEVICE'S COMMUNICATION.
- FAILURE OF THE HUB DISABLES THE ENTIRE NETWORK.





QUESTIONS

C. FIVE DEVICES ARRANGED IN A BUS TOPOLOGY:

FAILURE OF THE MAIN CABLE (BACKBONE) DISRUPTS THE ENTIRE NETWORK BECAUSE ALL DEVICES DEPEND ON IT.

D. FIVE DEVICES ARRANGED IN A RING TOPOLOGY:

FAILURE OF ONE CONNECTION BREAKS THE RING AND DISABLES THE ENTIRE NETWORK COMMUNICATION UNLESS A DUAL RING OR BYPASS MECHANISM IS USED.

- 7- • THIS IS A LAN (LOCAL AREA NETWORK) BECAUSE THE TWO COMPUTERS ARE CONNECTED WITHIN A SMALL GEOGRAPHICAL AREA (E.G., A HOME).
- LANS TYPICALLY INVOLVE HIGH-SPEED, LOW-COST CONNECTIONS WITHIN BUILDINGS OR SMALL AREAS, MATCHING THIS SETUP.

